

Stahl's
Essential Psychopharmacology

Prescriber's Guide

SEVENTH EDITION

Stephen M. Stahl



CAMBRIDGE

Medicine

Stahl's
Essential Psychopharmacology

Prescriber's Guide

SEVENTH EDITION

Stephen M. Stahl



CAMBRIDGE

Medicine

STAHL'S ESSENTIAL PSYCHOPHARMACOLOGY

Prescriber's Guide

SEVENTH EDITION

With the range of psychotropic drugs expanding and the usages of existing medications diversifying, we are pleased to present this very latest edition of what has become the indispensable formulary in psychopharmacology.

This new edition features several new compounds as well as information about new formulations, new indications, and new warnings for existing drugs.

With its easy-to-use, template-driven navigation system, the *Prescriber's Guide* combines evidence-based data with clinically informed advice to support everyone prescribing in the field of mental health.

Stephen M. Stahl is Professor of Psychiatry and Neuroscience at the University of California, Riverside and San Diego and Honorary Visiting Senior Fellow in Psychiatry at the University of Cambridge, UK. He has conducted various research projects awarded by the National Institute of Mental Health, Veterans Affairs, and the pharmaceutical industry. Author of more than 500 articles and chapters, Dr Stahl is also the author of the bestseller *Stahl's Essential Psychopharmacology*.

STAHL'S ESSENTIAL PSYCHOPHARMACOLOGY

Prescriber's Guide

SEVENTH EDITION

Stephen M. Stahl

*University of California at Riverside and at San Diego, Riverside and San
Diego, California*

Editorial assistant

Meghan M. Grady

With illustrations by

Nancy Muntner



CAMBRIDGE
UNIVERSITY PRESS



University Printing House, Cambridge CB2 8BS, United Kingdom

One Liberty Plaza, 20th Floor, New York, NY 10006, USA

477 Williamstown Road, Port Melbourne, VIC 3207, Australia

314-321, 3rd Floor, Plot 3, Splendor Forum, asola District Centre, New Delhi –110025,
India

79 Anson Road, #06–04/06, Singapore 079906

Cambridge University Press is part of the University of Cambridge.

It furthers the University's mission by disseminating knowledge in the pursuit of education,
learning, and research at the highest international levels of excellence.

www.cambridge.org

Information on this title: www.cambridge.org/9781316618134

© Stephen M. Stahl 2005, 2006, 2009, 2011, 2014, 2017, 2021

This publication is in copyright. Subject to statutory exception and to the provisions of relevant
collective licensing agreements, no reproduction of any part may take place without the written
permission of Cambridge University Press.

First published 2005

Revised and updated edition published 2006

Third edition published 2009

Fourth edition published 2011

Fifth edition published 2014

Sixth edition published 2017

Seventh edition published 2021

Printed in the United Kingdom by TJ Books Ltd, Padstow Cornwall

A catalog record for this publication is available from the British Library.

ISBN 978-1-108-92601-0 Paperback

ISBN 978-1-108-92602-7 Spiral

Additional resources for this publication at www.stahlonline.org

Every effort has been made in preparing this book to provide accurate and up-to-date information that is in accord with accepted standards and practice at the time of publication. Although case histories are drawn from actual cases, every effort has been made to disguise the identities of the individuals involved. Nevertheless, the authors, editors, and publishers can make no warranties that the information contained herein is totally free from error, not least because clinical standards are constantly changing through research and regulation. The authors, editors, and publishers therefore disclaim all liability for direct or consequential damages resulting from the use of material contained in this book. Readers are strongly advised to pay careful attention to information provided by the manufacturer of any drugs or equipment that they plan to use.

Cambridge University Press has no responsibility for the persistence or accuracy of URLs for external or third-party Internet Web sites referred to in this publication, and does not guarantee that any content on such Web sites is, or will remain, accurate or appropriate.

Contents

[Introduction](#)

[List of icons](#)

[1 acamprosate](#)

[2 agomelatine](#)

[3 alprazolam](#)

[4 amisulpride](#)

[5 amitriptyline](#)

[6 amoxapine](#)

[7 amphetamine \(d\)](#)

[8 amphetamine \(d,l\)](#)

[9 aripiprazole](#)

[10 armodafinil](#)

[11 asenapine](#)

[12 atomoxetine](#)

[13 benztropine](#)

[14 blonanserin](#)

[15 bremelanotide](#)

[16 brexanolone](#)

[17 brexpiprazole](#)

[18 buprenorphine](#)

[19 bupropion](#)

[20 buspirone](#)

[21 caprylidene](#)

[22 carbamazepine](#)

[23 cariprazine](#)

[24 chlordiazepoxide](#)

[25 chlorpromazine](#)

- [26 citalopram](#)
- [27 clomipramine](#)
- [28 clonazepam](#)
- [29 clonidine](#)
- [30 clorazepate](#)
- [31 clozapine](#)
- [32 cyamemazine](#)
- [33 desipramine](#)
- [34 desvenlafaxine](#)
- [35 deutetrabenazine](#)
- [36 dextromethorphan](#)
- [37 diazepam](#)
- [38 diphenhydramine](#)
- [39 disulfiram](#)
- [40 donepezil](#)
- [41 dothiepin](#)
- [42 doxepin](#)
- [43 duloxetine](#)
- [44 escitalopram](#)
- [45 esketamine](#)
- [46 estazolam](#)
- [47 eszopiclone](#)
- [48 flibanserin](#)
- [49 flumazenil](#)
- [50 flunitrazepam](#)
- [51 fluoxetine](#)
- [52 flupenthixol](#)
- [53 fluphenazine](#)

- [54 flurazepam](#)
- [55 fluvoxamine](#)
- [56 gabapentin](#)
- [57 galantamine](#)
- [58 guanfacine](#)
- [59 haloperidol](#)
- [60 hydroxyzine](#)
- [61 iloperidone](#)
- [62 imipramine](#)
- [63 isocarboxazid](#)
- [64 ketamine](#)
- [65 lamotrigine](#)
- [66 lemborexant](#)
- [67 levetiracetam](#)
- [68 levomilnacipran](#)
- [69 lisdexamfetamine](#)
- [70 lithium](#)
- [71 lofepramine](#)
- [72 lofexidine](#)
- [73 loflazepate](#)
- [74 lorazepam](#)
- [75 loxapine](#)
- [76 lumateperone](#)
- [77 lurasidone](#)
- [78 maprotiline](#)
- [79 memantine](#)
- [80 methylfolate \(l\)](#)
- [81 methylphenidate \(d\)](#)

[82 methylphenidate \(d,l\)](#)
[83 mianserin](#)
[84 midazolam](#)
[85 milnacipran](#)
[86 mirtazapine](#)
[87 moclobemide](#)
[88 modafinil](#)
[89 molindone](#)
[90 nalmefene](#)
[91 naltrexone](#)
[92 naltrexone/bupropion](#)
[93 nefazodone](#)
[94 nortriptyline](#)
[95 olanzapine](#)
[96 oxazepam](#)
[97 oxcarbazepine](#)
[98 paliperidone](#)
[99 paroxetine](#)
[100 perospirone](#)
[101 perphenazine](#)
[102 phenelzine](#)
[103 phentermine/topiramate](#)
[104 pimavanserin](#)
[105 pimozide](#)
[106 pipothiazine](#)
[107 pitolisant](#)
[108 prazosin](#)
[109 pregabalin](#)

[110](#) [_propranolol](#)
[111](#) [_protriptyline](#)
[112](#) [_quazepam](#)
[113](#) [_quetiapine](#)
[114](#) [_ramelteon](#)
[115](#) [_reboxetine](#)
[116](#) [_risperidone](#)
[117](#) [_rivastigmine](#)
[118](#) [_selegiline](#)
[119](#) [_sertindole](#)
[120](#) [_sertraline](#)
[121](#) [_sildenafil](#)
[122](#) [_sodium oxybate](#)
[123](#) [_solriamfetol](#)
[124](#) [_sulpiride](#)
[125](#) [_suvorexant](#)
[126](#) [_tasimelteon](#)
[127](#) [_temazepam](#)
[128](#) [_thioridazine](#)
[129](#) [_thiothixene](#)
[130](#) [_tiagabine](#)
[131](#) [_tianepetine](#)
[132](#) [_topiramate](#)
[133](#) [_tranylcypromine](#)
[134](#) [_trazodone](#)
[135](#) [_triazolam](#)
[136](#) [_trifluoperazine](#)
[137](#) [_trihexyphenidyl](#)

138 [triiodothyronine](#)

139 [trimipramine](#)

140 [valbenazine](#)

141 [valproate](#)

142 [varenicline](#)

143 [venlafaxine](#)

144 [vilazodone](#)

145 [vortioxetine](#)

146 [zaleplon](#)

147 [ziprasidone](#)

148 [zolpidem](#)

149 [zonisamide](#)

150 [zopiclone](#)

151 [zotepine](#)

152 [zuclopenthixol](#)

[Index by Drug Name](#)

[Index by Use](#)

[Index by Class](#)

[Abbreviations](#)

Introduction

This *Guide* is intended to complement *Stahl's Essential Psychopharmacology*. *Stahl's Essential Psychopharmacology* emphasizes mechanisms of action and how psychotropic drugs work upon receptors and enzymes in the brain. This *Guide* gives practical information on how to use these drugs in clinical practice.

It would be impossible to include all available information about any drug in a single work, and no attempt is made here to be comprehensive. The purpose of this *Guide* is instead to integrate the art of clinical practice with the science of psycho-pharmacology. That means including only essential facts in order to keep things short. Unfortunately it also means excluding less critical facts as well as extraneous information, which may nevertheless be useful to the reader but would make the book too long and dilute the most important information. In deciding what to include and what to omit, the author has drawn upon common sense and 30 years of clinical experience with patients. He has also consulted with many experienced clinicians and analyzed the evidence from controlled clinical trials and regulatory filings with government agencies.

In order to meet the needs of the clinician and to facilitate future updates of this *Guide*, the opinions of readers are sincerely solicited. Feedback can be emailed to customerservice@neiglobal.com. Specifically, are the best and most essential psychotropic drugs included here? Do you find any factual errors? Are there agreements or disagreements with any of the opinions expressed here? Are there suggestions for any additional tips or pearls for future editions? Any and all suggestions and comments are welcomed.

All of the selected drugs are presented in the same format in order to facilitate rapid access to information. Specifically, each drug is broken down into five sections, each designated by a unique color background: Therapeutics, Side Effects, Dosing and Use, Special Populations, and The Art of Psychopharmacology, followed by key references.

Therapeutics covers the brand names in major countries; the class of drug; what it is commonly prescribed and approved for by the United States Food and Drug Administration (FDA); how the drug works; how long it takes to work; what to do if it works or if it doesn't work; the best augmenting combinations for partial response or treatment resistance; and the tests (if any) that are required.

Side Effects explains how the drug causes side effects; gives a list of notable, life-threatening, or dangerous side effects; gives a specific rating for weight gain or sedation; and gives advice about how to handle side effects, including best augmenting agents for side effects.

Dosing and Use gives the usual dosing range; dosage forms; how to dose and dosing tips; symptoms of overdose; long-term use; if habit forming, how to stop; pharmacokinetics; drug interactions; when not to use; and other warnings or precautions.

Special Populations gives specific information about any possible renal, hepatic, and cardiac impairments, and any precautions to be taken for treating the elderly, children, adolescents, and pregnant and breast-feeding women.

The Art of Psychopharmacology gives the author's opinions on issues such as the potential advantages and disadvantages of any one drug, the primary target symptoms, and clinical pearls to get the best out of a drug.

In addition, drugs for which switching between medications can be complicated have a special section called The Art of Switching, which includes clinical pearls and graphical representations to help guide the switching process.

There is a list of icons used in this *Guide* following this Introduction and at the back of the *Guide* are several indices. The first is an index by drug name, giving both generic names (uncapitalized) and trade names (capitalized and followed by the generic name in parentheses). The second is an index of common uses for the generic drugs included in the *Guide* and is organized by disorder/symptom. Agents that are approved by the FDA for a particular use are shown in bold. The third index is organized by drug class and lists all the agents that fall within each particular class. In addition to these indices there is a list of abbreviations.

Readers are encouraged to consult standard references [1](#) and comprehensive psychiatry and pharmacology textbooks for more in-depth information. They are also reminded that the Art of Psychopharmacology section is the author's opinion.

It is strongly advised that readers familiarize themselves with the standard use of these drugs before attempting any of the more exotic uses discussed, such as unusual drug combinations and doses. Reading about both drugs before augmenting one with the other is also strongly recommended. Today's psychopharmacologist should also regularly track blood pressure, weight, and body mass index for most of his or her patients. The dutiful clinician will also check out the drug interactions of non-central nervous system (CNS) drugs with those that act in the CNS, including any prescribed by other clinicians.

Certain drugs may be for experts only, and these might include clozapine, thioridazine, pimozide, nefazodone, and monoamine oxidase (MAO) inhibitors, among others. Off-label uses not approved by the FDA and inadequately studied doses or combinations of drugs may also be for the expert only, who can weigh risks and benefits in the presence of sometimes vague and conflicting evidence. Pregnant or nursing women, or people with two or more psychiatric illnesses, substance abuse, and/or a concomitant medical illness may be suitable patients for the expert only. Controlled substances also require expertise. Use your best judgment as to your level of expertise and realize that we are all learning in this rapidly advancing field. The practice of medicine is often not so much a science as it is an art. It is important to stay within the standards of medical care for the field, and also within your personal comfort zone, while trying to help extremely ill and often difficult patients with medicines that can relieve their suffering and sometimes transform their lives.

Finally, this book is intended to be genuinely helpful for practitioners of psychopharmacology by providing them with the mixture of facts and opinions selected by the author. Ultimately, prescribing choices are the reader's responsibility. Every effort has been made in preparing this book to provide accurate and up-to-date information in accord with accepted standards and practice at the time of publication. Nevertheless, the psychopharmacology field is evolving rapidly and the author and publisher make no warranties that the information contained herein is totally free from error, not least because clinical standards are constantly changing through research and regulation. Furthermore, the author and publisher disclaim any responsibility for the continued currency of this information and disclaim all liability for any and all damages, including direct or

consequential damages, resulting from the use of information contained in this book. Doctors recommending and patients using these drugs are strongly advised to pay careful attention to, and consult information provided by, the manufacturer.

.....
¹For example, *Physician's Desk Reference* and *Martindale: The Complete Drug Reference* .

List of icons



agomelatine



alcohol dependence treatment



alpha adrenergic blocker



alpha 2 agonist



anticonvulsant



antiparkinson/anticholinergic



benzodiazepine



benzodiazepine receptor antagonist



beta blocker



cholinesterase inhibitor



dopamine 2 antagonist



dopamine 2 partial agonist



dual orexin receptor antagonist



flibanserin



histaminic



lithium



medical food



melanocortin receptor agonist



l-methylfolate



modafinil (wake-promoter)



monoamine oxidase inhibitor



naltrexone/bupropion



nefazodone (serotonin antagonist/reuptake inhibitor)



neuroactive steroid



nicotinic partial agonist



N-methyl-D-aspartate antagonist



noradrenergic and specific serotonergic antidepressant



norepinephrine and dopamine reuptake inhibitor



phentermine/topiramate



phosphodiesterase inhibitor



pimavanserin



sedative-hypnotic



selective norepinephrine reuptake inhibitor



selective serotonin reuptake inhibitor



serotonin-dopamine antagonist

=====



serotonin and norepinephrine reuptake inhibitor

=====



serotonin 1A partial agonist

=====



serotonin partial agonist reuptake inhibitor

=====



sodium oxybate

=====



stimulant

=====



thyroid hormone

=====



trazodone (serotonin antagonist/reuptake inhibitor)

=====



tricyclic/tetracyclic antidepressant

=====



vesicular monoamine transporter 2 inhibitor

=====



vortioxetine

=====



How the drug works, mechanism of action



Best augmenting agents to add for partial response or treatment resistance



Life-threatening or dangerous side effects



Weight Gain : Degrees of weight gain associated with the drug, with unusual signifying that weight gain has been reported but is not expected; not unusual signifying that weight gain occurs in a significant minority; common signifying that many experience weight gain and/or it can be significant in amount; and problematic signifying that weight gain occurs frequently, can be significant in amount, and may be a health problem in some patients



Sedation : Degrees of sedation associated with the drug, with unusual signifying that sedation has been reported but is not expected; not unusual signifying that sedation occurs in a significant minority; common signifying that many experience sedation and/or it can be significant in amount; and problematic signifying that sedation occurs frequently, can be significant in amount, and may be a health problem in some patients



Tips for dosing based on the clinical expertise of the author



Drug interactions that may occur



Warnings and precautions regarding use of the drug



Dosing and other information specific to children and adolescents



Information regarding use of the drug during pregnancy



Clinical pearls of information based on the clinical expertise of the author



The art of switching



Suggested reading

Acamprosate

Therapeutics

Brands

- Campral
- *see index for additional brand names*

Generic?

Not in USA

Class

- Neuroscience-based Nomenclature: glutamate multimodal (Glu-MM)
- Alcohol dependence treatment

Commonly Prescribed for

(bold for FDA approved)

- **Maintenance of alcohol abstinence**

How the Drug Works

- Theoretically reduces excitatory glutamate neurotransmission and increases inhibitory gamma-aminobutyric acid (GABA) neurotransmission

- Binds to and blocks certain glutamate receptors, including metabotropic glutamate receptors
- Because withdrawal of alcohol following chronic administration can lead to excessive glutamate activity and deficient GABA activity, acamprosate can act as “artificial alcohol” to mitigate these effects

How Long Until It Works

- Has demonstrated efficacy in trials lasting between 13 and 52 weeks

If It Works

- Increases abstinence from alcohol

If It Doesn't Work

- Evaluate for and address contributing factors
- Consider switching to another agent
- Consider augmenting with naltrexone



Best Augmenting Combos for Partial Response or Treatment Resistance

- Naltrexone
- Augmentation therapy may be more effective than monotherapy
- Augmentation with behavioral, educational, and/or supportive therapy in groups or as an individual is probably key to successful treatment